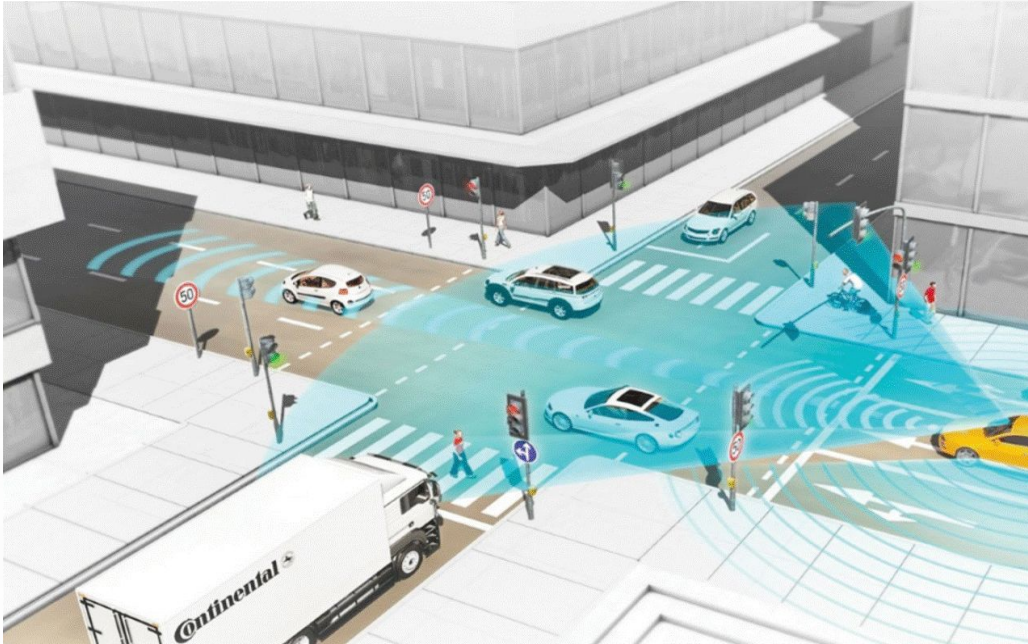


Detecting and Classifying Changes in Traffic Rules using Induction Loop Data



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Motivation

- Up-to-date maps are crucial for traffic safety.
- Detecting changes is challenging and costly
- Traffic flow to narrow down the search space

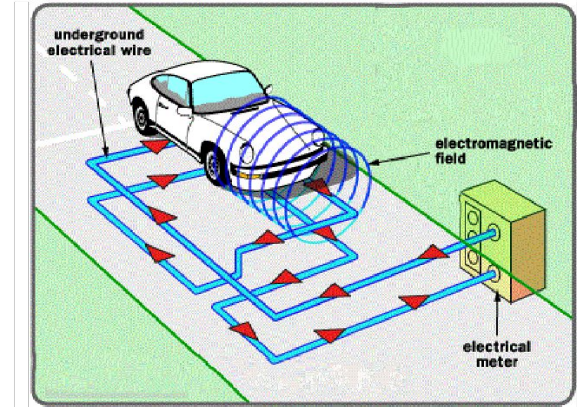


Changes (traffic rules)

Our work vs literature

- GPS from private vehicles are not available
- Public transport is not informative
- Videos - unrelated to traffic rules /complex
- Induction loop traffic flow data -
 - easily available
 - detect passing or stationary vehicles

Induction loop detector



Data modality	Event type	Classification
Trajectory	Festive events	No
Video	Car accidents, off- road driving, etc	No
Induction loop data (Ours)	Road blockage, traffic sign changes	Yes

Proposal

- DL method to detect and classify changes in traffic rules:
- Using induction loops.
- Seven types of changes:
 - road blockage
 - speed limit
 - U- turns
 -
- Simulated and real-world Luxembourg data
- Multi-class scenario



Luxembourg roadmap

Challenges

- Scarcity of induction loops results in **sparse data**
- High resolutions provide rich information but heavy/complex input size
- Class imbalance in the data (“no change” vs “change”)

s p a r s e

	7					6
	7	6	3		4	
	4	3				
4	2					
				3	2	4

DENSE

0	7	0	0	0	0	6
0	7	6	3	0	4	0
0	4	3	0	0	0	0
4	2	0	0	0	0	0
0	0	0	0	3	2	4



Contributions

Problem	Our solution
Data sparsity	• Multi-dimensional sliding windows • Residual connections • Multiple down-sampling
High dimensionality	• Dimensionality reduction • Stateful recurrency
Class imbalance	• Loss function with sample weighing
Additional improvement	• Data augmentation • Prediction weighing

Methodology (Part 1)

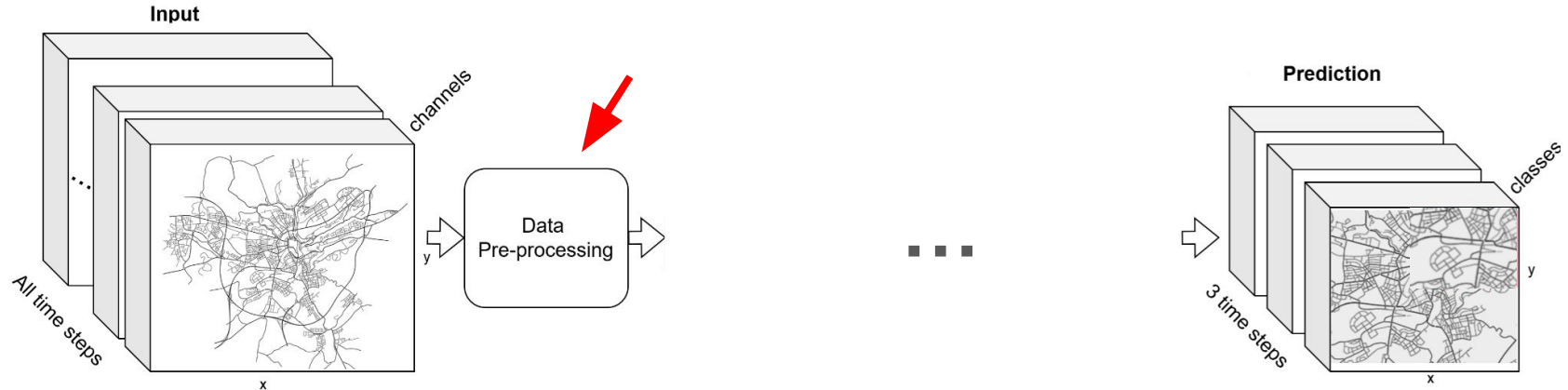
- 4D Input data (timestep, spatial x, spatial y, **channels**)
- **Channels**: average speed, road occupancy, road mask
- 4D Output data (timestep, spatial x, spatial y, class)



Methodology (Part 2)

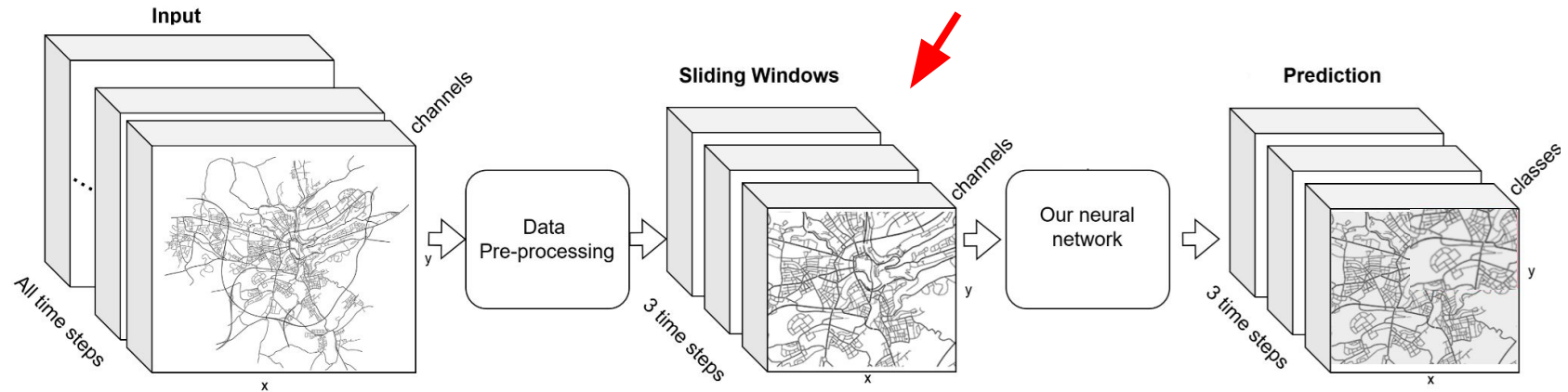
Pre-processing to reduce sparsity

- Down-sampling spatial dimensions
- Data augmentation with rotations



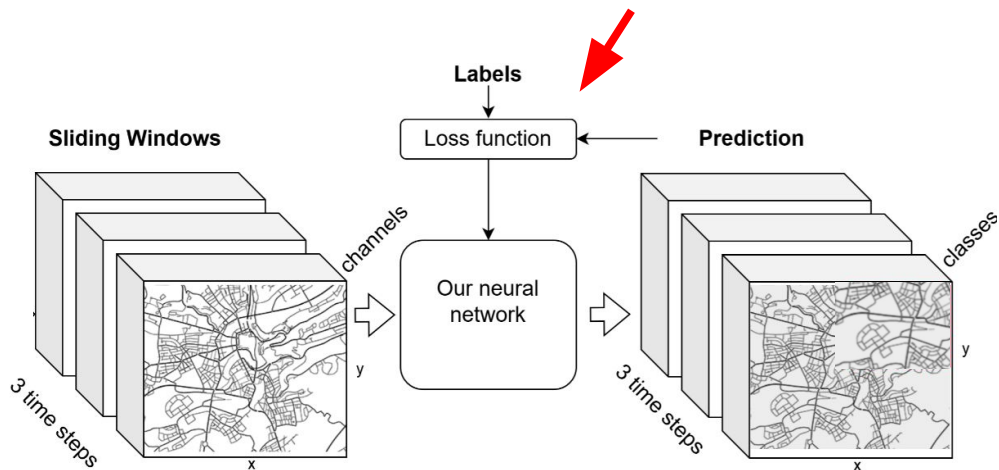
Methodology (Part 3)

- 3D sliding windows with overlaps
 - used as input for the neural network
 - reduces complexity of input
 - 1 window: 2.5km² and 6 minutes



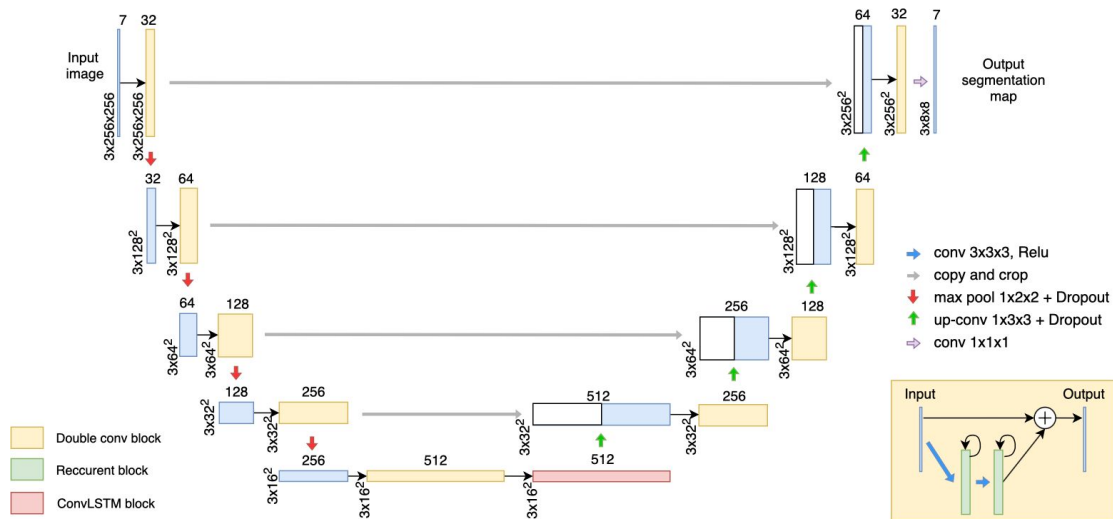
Methodology (Part 4)

- We use weighted cross-entropy loss function
- Loss weighting
 - Addresses class imbalance (“no change”)
 - Pseudo class balance



Methodology (Part 5)

- NN is based on U-Net (for semantic segmentation)
- Allows 4D input and 4D output
- Residual connections
- Stateful LSTM



Evaluation

We evaluated the performance:

- In simulated scenario -> detection and classification
- In real scenario -> only detection (unavailability of annotations)

Experiments: *Simulated Data and Ablation Studies*

- Tested on simulated data
- Baseline individually incorporated with some of the contributions.

Method	Class.	Detect.	Notes
<u>Baseline U-Net</u>	53.64%	79.05%	No contributions added
3D overlapping windows	62.45%	88.18%	Improves classification by 9%
Multiple down-sampling	66.46%	78.46%	Sparsity is reduced by 6%
Loss weighing	60.10%	85.97%	Combats class imbalance
Prediction weighing	61.74%	66.26%	Boosts classification score
<u>Complete solution</u>	76.18%	92.53%	All contributions added

Experiments: *Real-World Data*

- Change detection only

	Train set	Test set	F1-Score
	Simulated	Real-world	69.03%
	Simulated + real-world	Real-world	80.09%
	No real-world train set -> limited availability		

Limitations and Future Work

- Improved traffic data representation
- Explore the minimum volume
- Speed of change detection
- Identify propagation beginning
- Bridge the domain gap



Conclusion

- Induction loops for traffic rule change detection and classification
- F1-scores exceeding 80%

Thank you!
Any questions?



Ground truth

- The ground truth for the real-world data was extracted using HERE Incident API.
- This API provides information about road blockages due to traffic incidents.
- The information also includes the type and location of each traffic incident, status, start and end time
- Sometimes the reason is also specified (e.g., construction work or accident).

METHODOLOGY (Part 5)

One input has 6 minutes of data.

But we don't want to make the predictions on every input of 6 minutes to be the final predictions.

Hence, every 4 hours, we get a final prediction (using stateful LSTM memory).

The final prediction is a weighted mean of the 40 (10 timesteps per hour * 4 h)

Weights being defined by the road occupancy (more cars -> more weight).

Experiments: *Simulated Data and Ablation Studies (Part 2)*

Method	Class.	Detect.	Notes
Statefulness in LSTM	65.13	84.97	Captures longer-term temporal dependencies
Residual connections	59.46	79.65	Improves spatial and temporal dependencies
Data augmentation	53.65	79.65	Rotation transformation